

Generative AI in Special Education: Teachers' Insights on Instructional Enrichment vs. Accommodations

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Abstract

Generative Artificial Intelligence (GAI) holds significant potential for transforming education, yet its adoption in special education remains underexplored. It is important to investigate how special education teachers perceive and use GAI for two distinct instructional tasks: instructional enrichment and accommodations. Although the Technology Acceptance Model (TAM) has been extensively used to study technology adoption, its application to GAI in special education is underrepresented. This gap is critical as special education teachers must address diverse needs of students with multiple disabilities. Moreover, limited research had examined how TAM dimensions vary across different tasks. This late-breaking study revealed that teachers consistently rated GAI lower for accommodations across all TAM dimensions, with particularly differences in Intention to Use and Actual Use. The results highlighted the complexities of adopting GAI for distinct instructional purposes. Although GAI demonstrated potential for supporting instructional enrichment, its inability to generate individualized recommendations limited its effectiveness for accommodations.

CCS Concepts

• **Human-centered computing** → *Empirical studies in HCI*.

Keywords

Generative Artificial Intelligence, Special Education, Technology Acceptance Model, Instructional Enrichment, Instructional Accommodations

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1 Introduction

Generative Artificial Intelligence (GAI) refers to AI systems capable of creating content such as text, images, and other media by learning patterns from existing data [1]. In education, GAI has shown promise in enhancing teaching practices, particularly in areas requiring extensive personalization and creativity. Special education,

a field characterized by diverse and complex student needs, stands to benefit significantly from these advancements. Teachers in special education settings frequently manage tasks such as individualized instruction, curriculum adaptation, and classroom management, which contribute to high workloads and stress levels [8, 17]. GAI could alleviate some of these challenges by supporting teachers in two critical instructional tasks: enrichment and accommodations [11].

Instructional enrichment [19] involves enhancing the general curriculum through the creation of supplementary materials, activities, or resources that benefit the entire class. For instance, GAI can generate interactive exercises, contextual stories, or additional practice problems, helping teachers save time while improving instructional quality. However, the effectiveness of GAI in this domain depends on whether teachers perceive it as both useful and easy to integrate into their practices. On the other hand, instructional accommodations [2] require individualized adjustments tailored to specific students, such as alternative formats for assessments or step-by-step guides for students with disabilities. Although GAI can assist with content generation, its generalized outputs often fail to meet the highly specific needs of accommodations, posing unique challenges for adoption in this context.

This study explores how special education teachers perceive and use GAI for tasks with varying levels of complexity and purpose. By examining the differences in teachers' perceptions and usage patterns for enrichment and accommodation tasks, the study aims to provide insights into how GAI tools can be developed to better support the unique needs of special education teachers.

This study makes the following contributions. First, it expands the application of TAM to the context of special education, an area that has been underexamined in technology adoption research. Second, it underscores the importance of task-specific considerations, highlighting the need for tailored training and support to help teachers effectively integrate GAI into both instructional enrichment and accommodations tasks. Lastly, this study provides practical insights for improving GAI tools to better meet the needs of special education teachers, with the ultimate goal of alleviating their workload and enhancing educational outcomes for students with diverse learning needs.

2 Background and Related Work

2.1 Addressing Gaps in Technology Adoption Research

The Technology Acceptance Model (TAM) [6] is a foundational framework for understanding technology adoption, emphasizing two key factors: Perceived Usefulness (PU) and Perceived Ease of Use (PEU). These factors shape users' Intention to Use (IU), which

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subsequently predicts Actual Use (AU) [22]. In educational contexts, TAM has been extensively applied to analyze how teachers and students adopt digital tools. Research indicates that educators are more likely to integrate tools into their practices if they perceive them as useful for reducing repetitive tasks and improving content quality [12, 15, 22]. Despite growing interest in applying the TAM to technology adoption in education, significant gaps exist in its application to special education. Most research focuses on the use of general educational technologies to enhance accessibility and convenience for students or on the adoption of traditional technologies by special education teachers [23]. However, these studies rarely explore the distinct challenges and opportunities presented by advanced technologies like GAI. Additionally, it is unclear whether teachers' perceptions of GAI and their actual use of the technology vary between tasks, such as instructional enrichment and accommodations.

2.2 Task-Specific Adoption of GAI: Compare Instructional Enrichment and Accommodation

Prior research suggests that task characteristics can influence the perceptions and actual use of technology by highlighting the importance of aligning technological capabilities with the specific requirements of a task [22]. A closer alignment between task complexity and technological functionality leads to higher adoption rates and greater effectiveness [16]. Since tasks such as instructional enrichment and accommodations differ significantly in their demands and scope, it is important to compare teachers' perceptions and usage of technology across these tasks.

For instructional enrichment, GAI might align well with teachers' needs, potentially resulting in higher PU. Enrichment tasks often prioritize efficiency and broad applicability, areas where GAI might excel [14]. For example, GAI can quickly produce supplementary resources, examples, and explanations, enabling teachers to enhance the curriculum with minimal effort. Its ability to integrate seamlessly into existing workflows further reinforces its PU in this context. In terms of PEU, GAI might be particularly intuitive for enrichment tasks. These tasks typically involve straightforward outputs, such as generating examples or expanding existing content, which require simple prompts and minimal iterative adjustments. Teachers can readily adopt the tool without extensive training, making it highly appealing for enrichment purposes.

Conversely, instructional accommodations might pose unique challenges that may limit GAI's PU. Accommodations often require highly tailored and specific solutions to address the diverse needs of students with disabilities. GAI's tendency to produce generalized outputs might fail to meet these stringent personalization requirements, reducing its PU in this domain [27]. Similarly, GAI's PEU might be lower for accommodations due to the complexity of these tasks. Teachers must craft detailed, context-specific prompts and refine the outputs through iterative adjustments to ensure they are actionable and aligned with the needs of individual students [20]. The lack of intuitive or advanced features for creating highly customized solutions might further increase the cognitive and time burden, diminishing its usability for these tasks.

These differences in PU and PEU may directly influence teachers' IU for different tasks. For instructional enrichment, the combination of higher PU and PEU strengthens the teachers' intention to adopt GAI, as they perceive it as a valuable tool for enhancing instructional quality while reducing workload. Consequently, GAI is more likely to be integrated frequently and consistently for enrichment purposes. In contrast, for accommodations, lower PU and PEU weaken teachers' intention to use GAI. Although some teachers may recognize its potential, the significant effort required to generate actionable, individualized outputs can deter adoption. Teachers may perceive that the tool's limitations outweigh its benefits, particularly in the highly nuanced context of special education accommodations.

This study explores how GAI adoption varies between instructional enrichment and accommodations. Through this dual theoretical approach, it seeks to uncover actionable insights for designing GAI tools that better address the unique needs and challenges of special education teachers, ultimately enhancing adoption and effectiveness in these specialized teaching contexts.

3 Methods

This study employed an explanatory sequential mixed-methods design, where the quantitative phase informed the subsequent qualitative phase [4].

3.1 Participants

Participants were 52 special education teachers recruited from three schools. Data were collected through an online survey tool from October to November in 2024, with participants receiving a 5 US dollars incentive. Responses were filtered based on the completeness and quality, yielding an 82.54% valid response rate. Participants ranged from 20 to 45 years old ($Mean = 26.51$, $SD = 4.54$), with 28.8% male and 71.2% female. Popular GAI tools included chatGPT, Wenxinyiyan, and Kimi.

3.2 Quantitative Data Collection and Analysis Methods

The questionnaire began with demographic items, employing a nominal scale to gather information about participants' gender, age, teaching experience, subjects taught, prior engagement with GAI, and the specific types of GAI technologies they had used. The following section included a modified TAM questionnaire derived from existing related models [6, 24, 25].

3.2.1 Survey Design. The TAM questionnaire was designed to evaluate four key dimensions of GAI in instructional enrichment and accommodation. For instructional enrichment, PU assesses the perceived benefits of GAI, such as enhancing lesson quality and ensuring comprehensive curriculum coverage (e.g., "GAI-generated resources enhance the quality of instruction I provide to my students"). PEU evaluates how simple and effortless it is to integrate GAI into teaching, including its accessibility and ease of use (e.g., "GAI is easy to use for generating high-quality instructional resources"). IU captures educators' plans and commitment to incorporating GAI into their instructional practices (e.g., "I intend to use GAI regularly to create enriched teaching materials for my

students"). AU reflects the actual implementation of GAI, such as developing teaching materials and lesson plans (e.g., "I use GAI to generate teaching materials regularly").

For instructional accommodations, the focus shifts to GAI's role in supporting diverse learning needs. PU evaluates how GAI enhances accommodations by helping educators adapt content to suit varied student abilities and align with specific learning goals (e.g., "GAI helps me adapt instructional content to better accommodate students' diverse learning needs"). PEU measures the simplicity of using GAI to create personalized and accessible lesson materials (e.g., "GAI is easy to use for adapting instructional content to students' diverse learning needs"). IU captures educators' intentions to use GAI regularly to support differentiated instruction and address diverse needs (e.g., "I intend to use GAI to adapt instructional content for students with diverse learning needs"). Finally, AU examines how educators practically apply GAI to customize lesson plans and develop accommodations that enhance inclusivity and accessibility (e.g., "I use GAI to adapt instructional content for students with diverse learning needs").

3.2.2 Survey Refinement. The survey underwent rigorous refinement to ensure alignment with the study's objectives and theoretical framework. Prior to data collection, 20 GAI users and three academic experts completed the survey to provide feedback. The 20 GAI users, drawing on their extensive experience in special education, offered insights into the questionnaire's practicality, while the three academic experts with expertise in quantitative research methods assessed its structural coherence and validity. Based on their feedback, the survey was revised to enhance clarity in item phrasing, organization, layout, and numbering. This led to the final version of the 32-item TAM scale, with each dimension represented by four items across enrichment and accommodation. A seven-point Likert scale, ranging from "strongly disagree" (1) to "strongly agree" (7), was used to assess participants' perceptions and behaviors. To minimize response bias, the items were randomized in the questionnaire.

3.2.3 Quantitative Data Analysis. Data analysis was conducted using SPSS 23.0. First, the dataset was cleaned to prepare for analysis. Multivariate outliers were identified using Mahalanobis distances, while skewness and kurtosis analyses confirmed that the data exhibited a normal and symmetrical distribution, meeting the assumptions for subsequent statistical tests. To evaluate the reliability and validity of the measurement scales, internal consistency was assessed using *Cronbach's alpha* (α), ensuring robust reliability for each variable [5]. Additionally, confirmatory factor analysis was conducted to verify the validity of the scales, confirming their suitability for the study. Paired samples t-tests were then employed to analyze differences in PU, PEU, IU, and AU between instructional enrichment and accommodations tasks. Effect sizes were calculated using *Cohen's d* to quantify the magnitude of these differences, providing deeper insight into the practical significance of the findings.

3.3 Qualitative Data Collection and Analysis Methods

At the end of the survey, participants were invited to volunteer for follow-up interviews by providing their contact information. From the pool of volunteers, 5 participants were selected to ensure a range of diverse perspectives, considering factors such as years of teaching experience and familiarity with GAI tools. The semi-structured interviews were conducted online to accommodate participants' schedules and lasted between 35–70 minutes. These interviews followed a guide informed by prior research [7, 18], designed to explore educators' perceptions and experiences with GAI in the contexts of instructional enrichment and accommodations.

The interview guide covered a broad spectrum of topics, beginning with participants' professional background and experience with GAI tools. Key areas of focus included the practical applications of GAI for instructional tasks, perceived benefits and challenges, comparative advantages of GAI in enrichment versus accommodations, and broader opportunities and barriers to GAI adoption. For example, participants were asked to describe specific ways GAI tools enhance lesson personalization and accessibility, challenges encountered when implementing these tools, and any reservations about relying on AI for educational purposes. The interviews concluded with open-ended questions to allow participants to share additional insights or concerns regarding GAI usage in education.

Transcripts were analyzed using a widely recognized three-stage coding process [3]. Initial coding identified key concepts directly from the participants' responses, focused coding organized these concepts into coherent categories, and theoretical coding explored the relationships among categories to derive overarching themes. High inter-rater reliability was achieved (*Cohen's Kappa*, $\kappa = 0.79$), with discrepancies resolved through discussion among the researchers.

Thematic analysis, guided by an established framework [10], revealed three prominent themes, providing nuanced insights into how teachers perceive and use GAI for instructional enrichment and accommodations. These findings shed light on the opportunities and challenges associated with integrating GAI into diverse educational practices.

4 Results

4.1 Quantitative Results

A preliminary analysis confirmed a normal and symmetrical data distribution, allowing for rigorous evaluation of the measurement model's reliability and validity. Factor loadings were first calculated using a standard algorithm, with expected values meeting or exceeding the recommended threshold of 0.70. This threshold ensures that the indicators explain more than 50% of their variance, as suggested by prior study [21]. Next, internal consistency reliability was evaluated using α and composite reliability (CR). As presented in Table 1, both measures surpassed the 0.70 threshold, confirming the reliability of the items. Moreover, the model's convergent and discriminant validity were assessed using average variance extracted (AVE). The results show that all constructs had AVE values exceeding the 0.50 minimum, indicating strong convergent validity.

Table 1: Reliability and Convergent Validity of the Measurement Mode

Dimension	Construct	Item	Factor loading	Cronbach's α	Avg. variance extracted	Composite Reliability
Instructional Enrichment	PU	PU1	0.850	0.923	0.748	0.900
		PU2	0.932			
		PU3	0.734			
		PU4	0.797			
	PEU	PEU1	0.784	0.914	0.720	0.875
		PEU2	0.848			
		PEU3	0.845			
		PEU4	0.823			
	AU	AU1	0.898	0.932	0.778	0.893
		AU2	0.871			
		AU3	0.923			
		AU4	0.823			
	IU	IU1	0.855	0.905	0.753	0.887
		IU2	0.815			
		IU3	0.749			
		IU4	0.912			
Instructional Accommodations	PU	PU5	0.862	0.912	0.865	0.882
		PU6	0.884			
		PU7	0.834			
		PU8	0.873			
	PEU	PEU5	0.832	0.919	0.745	0.890
		PEU6	0.846			
		PEU7	0.875			
		PEU8	0.839			
	AU	AU5	0.923	0.970	0.846	0.893
		AU6	0.923			
		AU7	0.901			
		AU8	0.938			
	IU	IU5	0.878	0.914	0.834	0.902
		IU6	0.800			
		IU7	0.900			
		IU8	0.946			

Thus, the measurement model demonstrates adequate reliability and validity.

Table 2 presents the results of the paired samples t-test, including means, standard deviations, and *Cohen's d* effect sizes. Findings of the t-test revealed that teachers consistently rated GAI lower for accommodations compared to instructional enrichment across all TAM dimensions. However, the differences were particularly pronounced for IU and AU, both of which showed statistically significant disparities with large effect sizes. For AU, teachers reported significantly lower usage of GAI for accommodations (i.e., $t(51) = 7.83$, $p < 0.001$, $d = 1.086$), and for IU, GAI was also rated significantly lower for accommodations (i.e., $t(51) = 8.76$, $p < 0.001$, $d = 1.214$). Yet, differences in PU and PEU were smaller and not statistically significant. For PU, no significant difference was observed (i.e., $t(51) = 1.50$, $p = 0.140$, $d = 0.208$), and for PEU, the difference approached significance but did not reach the threshold (i.e., $t(51) = 1.80$, $p = 0.080$, $d = 0.250$).

4.2 Qualitative Results

The interview data revealed two central themes. First, teachers greatly valued GAI for its simplicity and speed in generating materials for instructional enrichment, describing it as intuitive and easy

to use, with no prior training required. Second, although GAI was helpful for enrichment, its recommendations for instructional accommodations were frequently criticized for being overly broad and impractical, often failing to address the specific needs of individual students effectively.

4.2.1 The Simplicity and Speed of Instructional Enrichment with GAI. One prominent theme was the ease and efficiency of using GAI for instructional enrichment. Participants described GAI as “simple”, “intuitive”, and “easy to get started” with no prior training required. For example, one of the special education teachers shared her experience with Wenxinyiyan: “I just provide a brief description of the topic, and GAI takes care of the rest. Sometimes, even a few keywords are enough. The materials it generates are diverse and tailored to my students’ needs, far surpassing what traditional resources can offer”. Time-efficiency was another key advantage highlighted by participants. Another participant with seven years of teaching experience noted: “Previously, I spent hours brainstorming and searching for ideas to make lessons engaging. Now, I can type a simple prompt, and within seconds, GAI produces innovative and varied materials”.

4.2.2 Broad and Impractical GAI Recommendations for Instructional Accommodations. While GAI was appreciated for enrichment tasks,

Table 2: Paired Samples t-Test Results: Means, Standard Deviations, and *Cohen's d*

Dimension	Instructional enrichment	Instructional accommodations	<i>t</i>	<i>p</i>	<i>d</i>
	Mean (Standard Deviation)	Mean (Standard Deviation)			
PU	5.855 (0.209)	5.750 (0.180)	1.50	0.140	0.208
PEU	5.720 (0.250)	5.680 (0.238)	1.80	0.080	0.250
AU	5.689 (0.344)	5.267 (0.250)	7.83	<.001	1.086
IU	5.800 (0.257)	5.650 (0.220)	8.76	<.001	1.214

its use for instructional accommodations presented significant challenges. All five teachers reported using GAI for accommodations, but three shared specific prompts that often led to unsatisfactory results. One common approach involved grouping students into general ability levels (e.g., A for higher ability, B for average ability, and C for lower ability) and requesting tiered objectives. For example, a teacher might use the prompt: "Please write the tiered teaching objectives for a *bubble-blowing* lesson, dividing students into A, B, and C levels." However, this broad grouping often led to impractical suggestions. One of the participants who used this method explained: "*The recommendations are often too generic. For example, GAI's suggestions for high-ability students are overly ambitious, while those for lower-ability students are still too challenging.*"

Some teachers tried to tackle this issue by providing detailed descriptions of their students' needs, such as their cognitive abilities, emotional tendencies, and behavioral traits. However, even with this effort, the results were disappointing. One experienced participant shared her perspective: "*I include specifics like a student's cognitive level, communication abilities, and past performance in my prompts, hoping for tailored advice. But the responses are still overly broad. For instance, I asked GAI for strategies to engage a student with severe disabilities in group activities. It suggested things like 'hands-on tasks' or 'sensory activities.' While those ideas aren't wrong, they're way too generic to be useful in my classroom.*"

5 Discussion

This study explored the perceptions and usage of GAI among teachers in special education for instructional enrichment and accommodations. Across all TAM dimensions, GAI was rated lower for instructional accommodations, with significant differences observed in IU and AU. These findings highlight the need to align GAI's capabilities with specific instructional requirements. While GAI struggles to provide the deep customization needed for accommodations, its strengths in simplifying enrichment tasks show its potential to reduce teacher workloads and improve educational outcomes in situations where broad, generalized solutions are sufficient, as noted by a previous study [11].

GAI is particularly well-suited for instructional enrichment, which focuses on enhancing general content rather than tailoring it to individual needs. Enrichment tasks benefit from GAI's ability to quickly generate diverse, high-quality materials, saving teachers time and improving the overall quality of instruction. This strong alignment explains the higher ratings and greater adoption of GAI for enrichment tasks, consistent with prior research [26]. Moreover, this finding resonates with the work [22], who revealed that teachers' adoption of digital technologies is strongly influenced

by the PU of the tool. In the case of enrichment tasks, GAI's efficiency in producing high-quality content enhances its PU, thereby driving adoption. Similarly, Granić and Marangunić [13] also highlight the importance of PEU in educational technology, which may explain why GAI is more readily embraced for enrichment tasks, where it is less complex to implement.

In contrast, GAI is less effective for accommodations due to its inability to address the nuanced needs of individual students. Its "one-size-fits-all" approach doesn't provide the level of personalization needed for unique learning profiles. Teachers found that GAI's outputs often lack the detail and customization required for accommodations. This highlights a mismatch between GAI's capabilities and the demands of creating effective accommodations. These findings align with the study from Farrokhnia et al. [9], which notes that GAI's strength in offering generalized responses limits its ability to deliver the personalized solutions needed in special education.

Furthermore, the study's findings regarding the varying adoption for enrichment and accommodation tasks align with the broader literature on technology adoption models. Scherer et al. [22] emphasize the importance of task-specific characteristics in influencing teachers' adoption decisions, highlighting that the perceived ease of use and usefulness of educational technology vary depending on the instructional task. In this study, GAI was seen as more useful and easier to use for enrichment tasks, reinforcing the idea that technology adoption is often influenced by the alignment between the tool's capabilities and the task demands.

5.1 Limitations and Future Research

This study has several limitations that should be acknowledged to better understand the findings and inform future research. First, the study did not have enough statistical power to thoroughly analyze differences in TAM pathways between instructional enrichment and accommodation. As a result, we were unable to investigate how perceptions of GAI differently influence its use for these two tasks. Future research should address this by utilizing larger samples and advanced modeling techniques to better examine these pathways. Second, the study was conducted within the context of special education in a single country, limiting the generalizability of the findings. Future studies should aim to include more diverse samples spanning different regions and educational systems to achieve a more comprehensive understanding of adoption patterns across various contexts. Finally, the qualitative findings in this study were not fully explored. Although participants identified key challenges with GAI, particularly in the context of instructional accommodations, the study did not provide enough detail on the specific rationales behind these difficulties. For example, teachers expressed frustration

with the generalized outputs, but the present study does not dive deeply into whether these challenges arise from the functionality of LLMs or from teachers' lack of familiarity with AI tools. This gap limits the ability to provide actionable recommendations for improving GAI tools to meet the specific needs of special education teachers.

6 Conclusion and Implications

This study highlights critical implications for integrating GAI into special education, particularly in the human-computer interaction (HCI). GAI's limitations in providing personalized accommodations underscore the need for tools that support nuanced individualization. Professional development should focus on training teachers to refine GAI outputs and integrate them with other resources, enabling tailored support for diverse student needs. Developers should prioritize customization capabilities, such as adaptive algorithms and interactive interfaces, to enhance GAI's ability to generate actionable, individualized recommendations. Task-specific tool design is essential to maximize GAI's effectiveness. While GAI excels in instructional enrichment by generating diverse, high-quality content efficiently, it struggles with deeply personalized accommodations. Recognizing these strengths and weaknesses ensures appropriate task-technology alignment. GAI adoption should therefore focus on leveraging its efficiency for enrichment tasks while addressing its limitations for accommodations through improved design and teacher support. These insights contribute to HCI by emphasizing the importance of aligning AI tools with specific educational needs, fostering more effective and equitable technology integration in special education.

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